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SECTION: 6A

COURSE: POWER ELECTRONICS

ASSIGNMENT #3

1) Several problems occur when a high-side MOSFET is driven by a 7406 open-collector inverter at frequencies higher than 30 kHz.

Such as:

- INSUFFICIENT GATE DRIVE VOLTAGE:

- i) Due to the 7406's open-collector output, the gate can only be pulled low and requires an external pull-up resistor to pull it high.

- ii) By limiting the gate drive voltage, the MOSFET may not be fully turned on, resulting in inefficient operation.

- SLOW SWITCHING SPEED:

- i) The pull-up resistor slows down the turn on time by introducing a time constant: $\tau = R_{\text{pull-up}} \times C_{\text{gate}}$.
- ii) At higher frequencies, this delay becomes significant, increasing switching losses.

- HIGH POWER DISSIPATION:

- i) Heat dissipation results from the MOSFET spending more time in the resistive region as a result of slow transition between the ON and OFF states.

- ii) This lowers efficiency and may require for more cooling methods.

- INCOMPATIBILITY WITH HIGH-SIDE DRIVER:

- i) In a high-voltage application, the 7406 is unable to directly supply the high voltage required to activate a high-side MOSFET.

- ii) Additional circuitry would be needed, such as a special high-side driver or a level shifter.

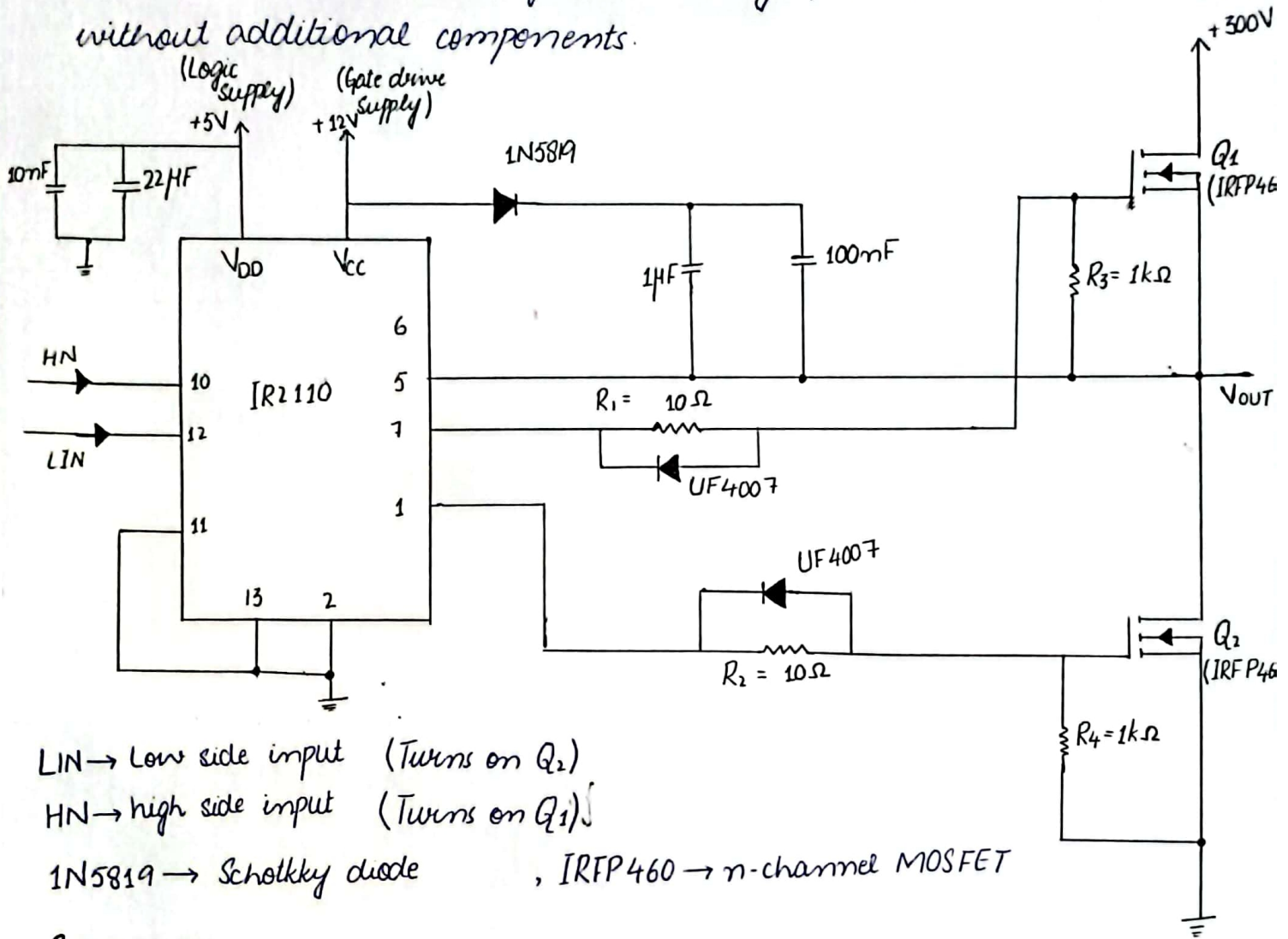
CONCLUSION:

Significant problems arise when a high-side MOSFET is driven by a 7406 open-collector inverter at high frequencies than 30 kHz. These problems include excessive power dissipation, slow switching speed, and inadequate gate drive voltage. The MOSFET's turn-on time is slowed by the pull-up resistor in 7406 output, which increases switching losses and lowers efficiency.

- 2) A special gate driver IC, such as IR2110, is a better option to deal with the above problems. The IR2110 is a high voltage, high speed power MOSFET with independent high and low side referenced output channels. It is designed for use in a wide variety of applications including motor control, power conversion, and power amplifiers. IR2110 offers the following:
 - HIGH-SIDE MOSFET DRIVER:
 - i) To ensure complete MOSFET turn on, the IR 2110 features an inbuilt bootstrap circuit (a technique used to generate a higher voltage needed to fully turn on a high-side MOSFET in a half bridge or a full-bridge driver) that supplies the required high-side driving voltage.
 - ii) The integrated circuit of the IR 2110 gate driver keeps in producing the additional voltage required to completely turn on a high-side MOSFET, enabling it to operate effectively.
 - FASTER SWITCHING SPEED:
 - i) The MOSFET gate capacitance is charged and discharged quickly due to special gate drivers which provide high peak current (e.g 12A for IR 2110)
 - LOWER POWER DISSIPATION:
 - i) Faster switching reduces the time spent in the resistive region, minimizing heat losses.

BETTER ~~LOGIC~~ CONTROL LOGIC:

- i) It can handle PWM signals directly from a microcontroller without additional components.



CONCLUSION:

To overcome the limitations of 7406, a gate driver, like IR2110 or TC4420 can be used. These drivers provide improved efficiency, increased gate drive voltage, and quicker switching. For instance, the IR2110 has a bootstrap circuit that enables appropriate high side operation, ensuring that the MOSFET turns completely ON and OFF.

3) FEATURE	7406 OPEN-COLLECTOR	IR2110 GATE DRIVER
Switching speed	Slow due to pull-up resistor	Fast due to high current drive
Power Dissipation	High due to slow transitions	Low due to rapid switching

FEATURE	7406 OPEN-COLLECTOR	IR2110 GATE DRIVER
Efficiency	Poor, significant losses	High, minimal heat generation
High-side capability	Requires extra circuitry	Built-in bootstrap support

IR2110 A BETTER CHOICE:

1) FASTER SWITCHING SPEED:

- The IR2110 gate driver provides high peak current (2A), allowing quick charging and discharging of the MOSFET gate capacitance.
- This reduces transition times, ensuring the MOSFET switches ON and OFF rapidly.

2) LOW POWER DISSIPATION:

- Faster switching reduces the time the MOSFET spends in the partially conducting region, minimizing power loss as heat.
- Less heat generation improves efficiency.

3) BETTER EFFICIENCY:

- Efficient gate drive ensures that the MOSFET operates either in fully ON (low resistance) or fully OFF (high resistance) state, minimizing losses.
- Improved efficiency is crucial for high-speed switching applications such as motor drivers and power converters.